

Matrix inequalities and norms

Chollet's permanent inequality for Hadamard products

Difficulty: extreme **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: The unrestricted PSD permanent inequality has resisted decades of work; it would advance a central collection of matrix-function inequalities.

For an $n \times n$ matrix $C = (c_{ij})$, define

$$\text{per } C = \sum_{\sigma \in S_n} \prod_{i=1}^n c_{i, \sigma(i)}.$$

Is it true that, for every integer $n \geq 1$ and every pair of complex Hermitian positive-semidefinite matrices $A, B \in \mathbb{C}^{n \times n}$,

$$\text{per}(A \circ B) \leq (\text{per } A)(\text{per } B), \quad (A \circ B)_{ij} = a_{ij}b_{ij}?$$

Relevance

The Hadamard product preserves positive semidefiniteness and corresponds to tensor products of Gram vectors. The conjecture seeks a computable structural upper bound for a difficult matrix polynomial under this basic matrix operation. It concerns matrix inequalities and permanent computation.

References

- J. Chollet, *Is there a permanent analogue to Oppenheim's inequality?*, American Mathematical Monthly 89 (1982), 57–58 ([original paper](#)).
- P. Pant and R. Singh, *On Chollet's Permanent Conjecture for Graph Laplacians*, arXiv:2604.24192 (2026), abstract and introduction, with structured partial results ([primary manuscript](#)).
- Q. Li, Q. Zhao, and W. Luo, *Chollet's Permanent Conjecture Through Order Six via Border Induction*, posted 31 August 2026, Theorem 1.1 ([primary preprint](#)).
- K. Rodtes, *Chollet's permanent conjecture for 4×4 matrices*, Linear and Multilinear Algebra 72(16) (2024), 2633–2638. [Published paper](#); [preprint](#).
- I. M. Wanless, *Lieb's permanent dominance conjecture* (2022), Conjecture 5 and the implication diagram distinguishing it from disproved stronger conjectures ([primary manuscript](#)).

Status check — 2026-09-10

The August 2026 preprint claims the displayed inequality through order six. That claim has not been independently refereed here and does not claim the unrestricted result; the unresolved target includes orders $n \geq 7$. The April 2026 paper addresses structured classes. Searches for “Chollet conjecture”, “permanent”, “proof”, and 2025–2026 found no general resolution. Disproofs of the stronger Bapat–Sunder and permanent-on-top conjectures do not refute this statement.

Audit update (2026-09-10): Rechecked the graph-Laplacian paper, the order-six preprint, and Rodtes's order-four result, and searched for general resolutions. Added the published order-four reference to distinguish established low-order progress from the newer unrefered order-six claim. This is a bounded literature check, not a proof that no solution exists.